

Navigating contradictions: *Salmonella* Typhimurium chemotactic responses to conflicting effector stimuli

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eLife Assessment

In this manuscript, Franco and colleagues describe **valuable** findings about the chemotactic response of *Salmonella* to serine and indole, conflicting chemotactic signals. Although the evidence presented is **solid**, concerns were raised about the novelty of the chemotactic phenomena observed with these two compounds. Also, although the induction of invasion by feces is a novel and interesting finding, the lack of follow-up to this observation was also noted.

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Abstract

Chemotaxis controls motility and colonization in many enteric pathogens, yet most studies have examined bacterial responses to single effectors in isolation. Previously, we reported that *Salmonella* Typhimurium uses the chemoreceptor Tsr to detect L-serine (L-Ser) in human blood serum, promoting invasion of damaged vasculature (Glenn *et al.*, *eLife* 2024 ¹). Tsr also mediates sensing of indole, a microbiota-derived chemorepellent and bactericide proposed to protect against enteric infection by deterring pathogen colonization. The major biological reservoir of indole in the gut is feces, where it accumulates to millimolar levels. Here, we tested whether indole-rich human fecal material is protective against infection and found that exposure to feces instead enhances intestinal invasion in an explant model. Surprisingly, diverse non-typhoidal *Salmonella* serovars were strongly attracted to feces despite its high indole content. We found that while pure indole is a strong repellent sensed through Tsr, its effects are overridden in the presence of nutrient attractants, including L-Ser. Moreover, indole only minimally impairs growth in the presence of sufficient nutrients. Using video microscopy, we observed that Tsr integrates L-Ser and indole signals in real time, biasing bacterial movement based on the relative concentrations of attractant and repellent. We propose that this chemotactic compromise optimizes pathogen fitness by guiding bacteria to niches with a favorable L-Ser-to-indole ratio, balancing nutrient acquisition and avoidance of high microbial competitor density. These findings highlight the limitations of single-effector

studies in predicting bacterial navigation in complex environments, where chemotaxis is shaped by the integration of multiple, often opposing, chemical cues.

Introduction

Many bacteria that colonize the gastrointestinal tracts of humans and other animals employ chemotaxis to sense chemical effectors in the gut lumen and swim to environments conducive for growth and colonization^{2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100}. This process is controlled by chemoreceptor proteins, which recognize chemical effectors and transduce signals through a phosphorylation cascade to regulate flagellar rotation and swimming direction, ultimately determining the spatial and temporal patterns of bacterial colonization (**Fig. 1A**)^{2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100}. In turn, the colonization topography of bacteria within the gut influences the health of the host through nutrient absorption, developmental regulation, and resistance to pathogens^{9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100}. While many effectors have been studied and characterized in isolation as attractants or repellents^{5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100}, natural environments contain mixtures of opposing signals. In the enteric lumen, bacteria encounter a complex milieu of conflicting effector stimuli, including nutrients, molecular cues, and toxins^{2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100}. Only a handful of studies have investigated how bacteria navigate conflicting chemical gradients and there remains much to learn about how bacteria prioritize these signals to direct their movement and colonization (**Fig. 1B**)^{2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100}.

A chemical effector of major importance for enteric bacterial communities is indole, an interbacterial signaling molecule that regulates diverse aspects of physiology and lifestyle^{20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100}. Indole is excreted by gut microbiota as a byproduct of tryptophan metabolism and accumulates to millimolar levels in human feces (**Fig. 1A-B**)^{20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100}. Indole is amphipathic and can transit bacterial membranes to regulate biofilm formation and motility, suppress virulence programs, and exerts bacteriostatic and bactericidal effects at high concentrations^{20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100}. Indole was one of the earliest identified chemorepellents, and subsequent work has extensively explored its role in *Escherichia coli* chemotaxis (Table S1)^{12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100}. Recent studies have advanced our understanding of the molecular mechanisms underlying *E. coli* indole taxis and the involvement of the chemoreceptor Tsr to serine and repellents (Tsr) (**Fig. 1A-B**)^{12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100}. From this body of research, the hypothesis emerged that indole from the gut microbiota repels pathogens and restricts their growth as a mechanism of colonization resistance conferred by the microbiome^{12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100}. If true, this could represent an interesting avenue for cultivating healthy microbiomes more robust against pathogen infiltration. However, indole taxis has only been studied in *E. coli* and no prior work has tested whether human fecal material, the major biological source of indole in the gut, induces frank pathogen chemorepulsion or inhibits pathogen growth at physiologically relevant levels (Table S1).

We were interested in further studying the role of Tsr in navigating contradictory effector stimuli because this chemoreceptor possesses an interesting dual function: it senses chemorepellents and also the amino acid and nutrient L-Ser as a chemoattractant (**Fig. 1A-B**)^{1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100}. Tsr plays important roles during enteric invasion^{33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100}, but questions remain about how the chemoreceptor regulates pathogen colonization in the presence of conflicting signals. We recently mapped the biological distribution of Tsr and reported that many enteric pathogens and pathobionts possess Tsr orthologues, including the genera *Salmonella*, *Citrobacter*, and *Enterobacter*^{1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100}. Whether these and other bacteria respond to indole as a chemorepellent has remained unclear because all prior studies of indole taxis focused on *E. coli* (Table S1). Prior *in vitro* work with purified effectors showed when both attractants and repellents are present, Tsr facilitates so-called “intermediate” responses between chemoattraction and chemorepulsion, suggesting that, in some cases, navigating conflicting stimuli is regulated at the single chemoreceptor level^{13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100}. These types of chemotactic compromises, such as those that have been referred to as bet-hedging, trade-off, and simultaneous sensing, remain insufficiently

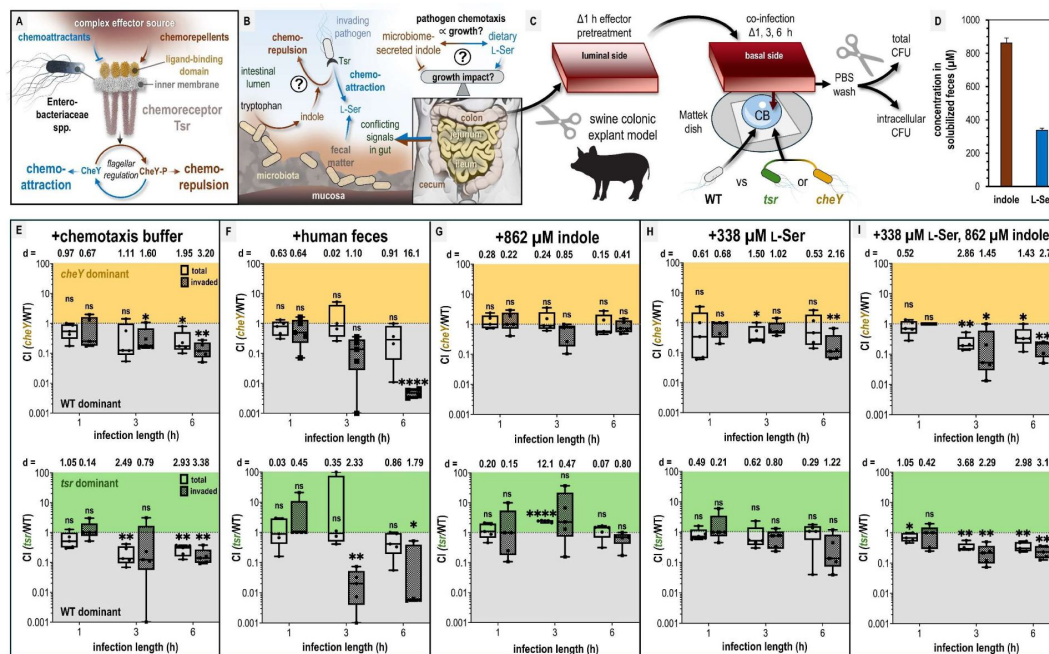


Fig. 1.

Tsr and chemotaxis mediate efficient pathogen invasion of colonic tissue.

A-B. Overview of the role of Tsr in chemotactic responses and premise of this study. C. Experimental design of colonic explant infections. See Materials & Methods for experiment details such as tissue dimensions. D. Serine (presumed to be nearly 100% L-Ser, see Materials & Methods) and indole content of liquid human fecal treatments, as measured by mass spectrometry. E-I. Competitive indices (CI) of colony-forming units (CFUs) recovered from co-infected swine explant tissue, either from the total homogenate (open box and whiskers plots), or from tissue washed with gentamicin to kill extracellular and attached cells, which we refer to as the “invaded” intracellular population (checked box and whisker plots), as indicated. Each data point represents a single experiment of a section of tissue infected with bacteria, and the CI of CFUs recovered from that tissue ($n=5$). Boxes show median values (line) and upper and lower quartiles, and whiskers show max and min values. Effect size (Cohen’s d) and statistical significance are noted for each experiment in relation to competitive advantage, i.e. deviation from a CI of 1 (not significant, ns; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$). See also Figure S1 for disaggregated CFU enumerations for each experimental group prior to CI calculation. Data S1 contains all numerical CFU measurements.

studied, but could be relevant to enteric infections because the gut lumen contains conflicting effectors ^{13,18,19}. Earlier work posited that bacteria sum conflicting signals and ultimately perform attraction or repulsion responses based on the relative concentrations and sensitivity to the individual effectors, which, if true, would enable straight-forward predictions for behaviors to effector mixtures ^{15,18}. But, this binary model of chemotaxis seems too coarse to explain recent observations of population structures that emerge in response to conflicting signals that are clearly distinct from either attraction or repulsion ^{13,19}.

In this study, we aimed to gain a deeper understanding into how chemotaxis directs bacterial localization to naturally-occurring stimuli that contain conflicting effectors, the types of population structures that arise and their dynamics of formation, and their potential relevance in enteric infections. *Salmonella enterica* serovar Typhimurium is a frank pathogen that relies on chemotaxis for enteric infection and employs an array of strategies to compete with the native microbiota for nutrients and colonization niches, ultimately invading the intestinal tissue and becoming intracellular ^{9,33–38}. *S. Typhimurium* also differs fundamentally from *E. coli* because it lacks tryptophanase genes and cannot itself produce indole, and thereby could provide a novel perspective on indole taxis ^{39,40}. In this study, we used the pathogen *S. Typhimurium* as a model system to: (1) test the hypothesis that the microbiota secretion product indole is protective against enteric invasion, (2) investigate if pathogens are repelled by indole-containing fecal material, and (3) learn how the chemoreceptor Tsr regulates pathogen spatial localization in response to conflicting chemotactic stimuli of the intestinal environ. Using live imaging, we directly visualized how enteric pathogen populations dynamically restructure in response to physiological mixtures of attractants and repellents. We demonstrate that chemotaxis responses to natural biological combinations of effectors, and their impacts on infection outcomes, are not easily predicted based on responses to individual effectors alone.

Results

Impact of opposing chemoeffector stimuli on pathogen invasion in swine colonic explants

We sought to test whether indole in human fecal matter protects against *S. Typhimurium* infection and whether this involves chemorepulsion mediated by the chemoreceptor Tsr. *S. Typhimurium* preferentially invades tissue of the distal ileum, but also infects the cecum and colon ^{41–44}. We presume the amount of indole is greatest in the lower gastrointestinal tract where tryptophan has been maximally digested by microbiota tryptophanases, and so we developed a swine colonic explant model that mimics the architecture and size of adult human colonic tissue ^{40,45–49}. We based this model on a prior study of explant tissue used to characterize invasion through gentamicin washes that kill extracellular and attached bacteria ⁵⁰; gentamicin washing is also a commonly-used method to quantify *S. Typhimurium* intracellular populations in cell culture experiments ^{51,52}. A section of colonic tissue was prepared for each experiment with gentle cleaning and treated with various effector solutions: solubilized human feces, purified indole and/or L-Ser at fecal-relevant concentrations, or buffer as a control (see Materials & Methods for tissue dimensions and other experiment details). This experimental setup simulates a biological gradient in which the effector concentration is initially highest near the tissue and diffuses outward into the buffer solution.

Following treatment with effectors, the tissue was subsequently exposed to co-infection with wild-type (WT) *S. Typhimurium* strain IR715 and either a *cheY* mutant (motile but non-responsive to chemoeffector stimuli) or *tsr* deletion mutant (**Fig. 1C-D**, **Table 1**) ³³. The functionality of these mutants have been confirmed through *in vivo* infection studies using genetically complemented strains ^{33,34}. To assess the role of chemotaxis in infection, we quantified total bacteria harvested from tissue homogenates and also enumerated bacteria following a gentamicin

wash, which we presume to reflect bacteria that were intracellular and evaded killing by the gentamicin wash, at 1, 3, and 6 h post-infection (**Fig. 1C**, Materials & Methods) ^{53,54}. For buffer-treated explant experiments, WT *S. Typhimurium* showed a modest time-dependent advantage in colonization and invasion compared to chemotactic mutants, indicating that chemotaxis, and specifically Tsr, promotes tissue colonization under baseline conditions (**Fig. 1E**, Fig. S1A-B, Data S1).

The hypothesis that indole protects against pathogen colonization predicts that feces, the major biological repository of gut indole, will protect against invasion. Contrary to this, we found that fecal treatment significantly enhances intracellular invasion of WT *S. Typhimurium* (compared to buffer treatment), providing a >100-fold competitive advantage mediated by Tsr (**Fig. 1F**, Fig. S1C-D). Analysis of the liquified human fecal matter used in this study revealed an indole concentration of 862 μ M, consistent with previously reported ranges (0.5–5 mM) (**Fig. 1D**, Materials & Methods) ^{20,23–26}. However, when colonic tissue was treated with purified indole at the same concentration, the competitive advantage of WT over the chemotactic mutants was abolished compared to fecal-treated tissue (**Fig. 1G**, Fig. S1E-F, Data S1). Given that Tsr mediates attraction to L-Ser for both *E. coli* and *S. Typhimurium*, we hypothesized that L-Ser present in feces might be responsible for increased colonization of fecal-treated tissue ^{1,2,55,56}. However, treatment with 338 μ M L-Ser, the concentration present in our fecal sample (**Fig. 1D**, Materials & Methods), actually shows a decrease in WT advantage compared to fecal treatment, similar to the effect of indole alone (**Fig. 1H**, Data S1). While treatments with either indole or L-Ser decrease the WT advantage, total bacterial loads are similar, suggesting that neither effector is “protective” in terms of infection burden (Fig. S1, Data S1). Since both effectors individually alter invasion, but do not on their own recapitulate the efficient invasion phenotype observed with fecal treatment, we then wondered whether the combination of indole and L-Ser might be sufficient. Interestingly, we found that treatment with a mixture of indole and L-Ser effectively neutralized their individual effects, resulting in a response nearly identical to buffer-treated tissue, and the WT advantage observed with fecal treatment was not recapitulated (**Fig. 1I**, Fig. S1I-J, Data S1).

These results provide insights into the relationship between chemotactic sensing of fecal material and its role in invasion, while also revealing complex behaviors that raise further questions. The key takeaway is that although pure indole reduces WT colonization advantage in a chemotaxis- and Tsr-dependent manner, indole within feces does not elicit the same response, nor does indole-rich fecal material provide protection against invasion in this model (**Fig. 1**, Fig. S1). This may be explained by our observation that L-Ser can cancel out the effects of indole (**Fig. 1G**, **Fig. 1I**, Data S1). These findings suggest that the presence of multiple chemical signals within complex biological effector sources can make it challenging to predict chemotactic behavior based on how the bacteria respond to singular effectors.

Curiously, when L-Ser and indole are present together, the magnitude of the WT advantage over the *cheY* and *tsr* mutants resembles a null response (i. e. similar to buffer treatment), rather than providing an additive fitness advantage like occurs with fecal treatment (**Fig. 1**, Data S1). One possible explanation is that fecal material contains other metabolites that influence sensing and growth ^{2,8,57–61}. Additionally, the WT advantage in the presence of fecal material could relate to Tsr-mediated energy taxis ^{33,34}, swimming regulation that encourages invasion ³⁸, or experimental treatment conditions that may impact bacterial growth. To further investigate these behaviors, the following sections examine chemotactic responses to these treatments and their effects on growth.

<i>Strain</i>	<i>Reference/Source</i>
<i>S. enterica</i> Typhimurium IR715 nalidixic acid derivative of ATCC 14028	Rivera-Chávez, F. et al. ³³
<i>S. enterica</i> Typhimurium IR715 $\Delta cheY::Tn10$ (Tet ^R)	Rivera-Chávez, F. et al. ³³
<i>S. enterica</i> Typhimurium IR715 $\Delta tsr::pFR3$ (Cm ^R)	Rivera-Chávez, F. et al. ³³
<i>S. enterica</i> SARA1	Beltran, P. et al. ⁷¹
<i>S. enterica</i> Newport M11018046001A	Shariat, N. et al. ⁶²
<i>S. enterica</i> Enteriditis 05E01375	Shariat, N. et al. ⁶²
<i>E. coli</i> BL21-DE3 Cat# 70954-3	Millipore-Sigma

Table 1

Bacterial Strains

Non-typhoidal *S. enterica* are attracted to human feces despite high indole content

Having found that Tsr mediates efficient colonic invasion for *S. Typhimurium* in our explant system when exposed to fecal material, we next sought to learn what chemotactic behaviors this chemoreceptor orchestrates in response to feces; given that this represents the highest concentration of indole *S. Typhimurium* might natively encounter, we expected to observe chemorepulsion (**Fig. 1D** [12](#), [13](#), [18](#), [27](#), [32](#)). We employed the chemosensory injection rig assay (CIRA) for live-imaging of bacterial chemotaxis responses to a source of effectors injected through a glass microcapillary [1](#). The flow and dynamic nature of the gut lumen make this a suitable approach for modeling and studying enteric chemotaxis responses *in vitro* [1](#).

In this assay, chemoattraction is observed as an influx of cells toward the effector source and chemorepulsion as decreasing cells (Figure S2). As described previously, effector injection introduces a steep microgradient, and, using mathematical modeling of the diffusion of the fecal sources of indole and L-Ser, we can approximate the local concentrations experienced by bacteria at a given distance from the injection source, which for most of the field of view is in the picomolar to low nanomolar range (**Fig. 2A** [1](#), Materials & Methods) [1](#). Using experiments with fluorescent dye we previously confirmed our calculated local effector concentrations to be in the range of 5-10% within 300 μm of the injection source [1](#). Further, we showed using a set of controls of buffers at different pH, and chemotaxis mutants, that the low flow-rate of our apparatus (approximately 300 femtoliters per minute) does not itself meaningfully perturb the bacterial population relative to the much larger shifts that occur due to chemotaxis responses [1](#).

Over five minutes, we found both WT and *tsr* exhibit strong chemoattraction to fecal material, whereas *cheY* remains randomly distributed (**Fig. 2B** [1](#), Movie 1). By examining the radial distribution of the bacterial populations, we found WT more tightly centers around the treatment source than *tsr* (**Fig. 2C-E** [1](#), Movie 1). In terms of the rate of bacterial accumulation, the chemoattraction of *tsr* lags behind the WT for the first 120 s (**Fig. 2F-G** [1](#), Movie 1). We wondered how these deficiencies in fecal attraction might translate to direct competition, where different strains are experiencing the same treatment source simultaneously. To address this, we performed CIRA with solubilized human feces and two strains present in the same pond, which we tracked independently through fluorescent markers (**Fig. 3** [1](#)). As expected, WT shows a strong chemoattraction response versus *cheY* (**Fig. 3A** [1](#), Movie 2). Interestingly, we found that when competed directly, WT vastly outperforms *tsr*, with the maximal bacterial distribution in proximity to the treatment source higher by about 4-fold (**Fig. 3B** [1](#), Movie 2). These data confirm that despite its high indole content, *S. Typhimurium* is attracted to human fecal material through chemotaxis, and this response involves Tsr, although not as the sole mediator. We expect the attraction of the *tsr* mutant is explained by the fact that *S. Typhimurium* possesses other chemoreceptors that detect glucose, galactose, and L-Asp as chemoattractants, which are present in human feces [2](#), [8](#), [57](#)–[61](#).

Recent work highlights how genetic diversity among strains and differences in Tsr expression, even within isogenic populations, modulate chemotaxis function [13](#), [35](#). To gain a broader perspective on fecal taxis, we examined the responses among diverse *Salmonella* serovars and strains responsible for human infections. Using dual-channel imaging, we compared *S. Typhimurium* IR715 with a clinical isolate of *S. Typhimurium*, SARA1, and found both strains exhibit attraction to feces, although SARA1 shows a slightly weaker response (**Fig. 3C** [1](#), Movie 3). We then tested a clinical isolate of *S. Newport*, an emerging cause of salmonellosis in the United States and Europe [62](#), [63](#). This strain is strongly attracted to fecal material, with a tighter accumulation of cells at the treatment source than *S. Typhimurium* IR715 (**Fig. 3D** [1](#), Movie 3). Lastly, we examined a clinical isolate of *S. Enteritidis*, a zoonotic pathogen commonly transmitted

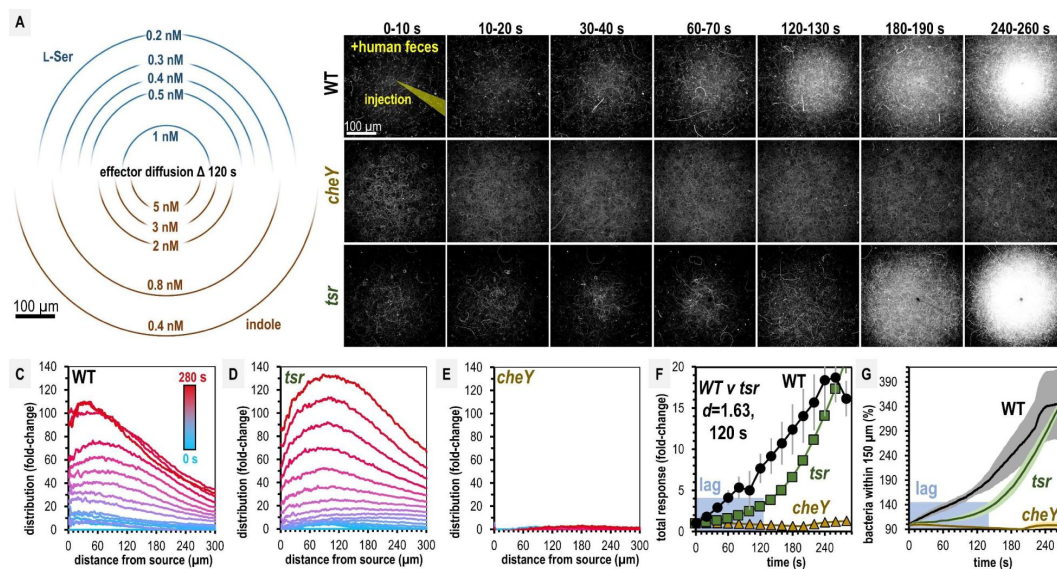


Fig. 2.

***Salmonella* Typhimurium exhibits attraction toward liquid human fecal material.**

A. Diffusion modeling showing calculated local concentrations in CIRA experiments with liquid human fecal material based on distance from the central injection source. B. Max projections of representative *S. Typhimurium* IR715 responses to a central source of injected liquid human fecal material. C-E. Bacterial population density over time in response to fecal treatment. The initial uniform population density in these plots is indicated with the blue line (time 0), and the final mean distributions with the red line (time 280 s), with the mean distributions between these displayed as a blue-to-red spectrum at 10 s intervals. F-G. Temporal analyses of area under the curve (AUC) or relative number of bacteria within 150 μm of the source. Effect size (Cohen's d) comparing responses of WT and *tsr* attraction at 120 s posttreatment is indicated. Data are means and error bars are standard error of the mean (SEM, $n=3-5$). See also Movie 1, Table S1, and Figure S2.

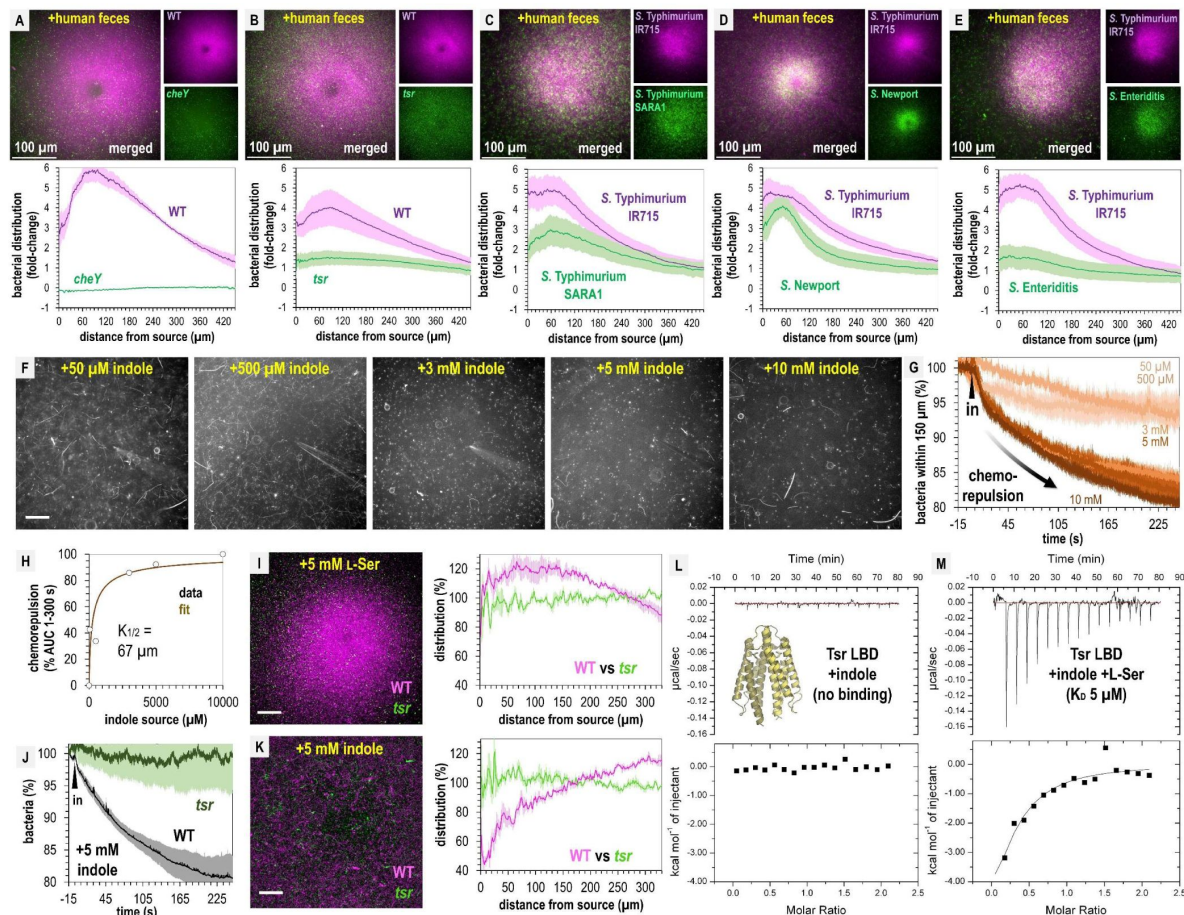


Fig. 3.

Fecal indole is insufficient for chemorepulsion but indole in isolation is a strong chemorepellent.

A-E. Dualchannel imaging of chemotactic responses to solubilized human feces by WT *S. Typhimurium* IR715 (pink) and isogenic mutants or clinical isolate strains (green), as indicated. Shown are max projections at time 295-300 s posttreatment. Data are means and error bars are standard error of the mean (SEM, $n=3-5$). See also Movies 2-3. F. Representative max projections of responses at 295-300 s of indole treatment. G-H. Quantification of chemorepulsion as a function of indole concentration ($n=3-5$). I-K. Comparison of WT and *tsr* mutant responses to L-Ser or indole. See also Fig. S2. L-M. Isothermal titration calorimetry (ITC) experiments with 50 μM *S. Typhimurium* Tsr ligand-binding domain (LBD) and indole, or with L-Ser in the presence of 500 μM indole. Data are means and error bars are standard error of the mean (SEM, $n=3-5$). AUC indicates area under the curve. Scale bars are 100 μm. See also Movies 2-3.

from poultry, which displays weak attraction to fecal material (Fig. 3E, Movie 3). Overall, we found that chemoattraction to fecal material is conserved among diverse non-typhoidal *Salmonella* serovars responsible for human infections, although the degree of attraction varies. Notably, despite the high indole content in feces, none of the strains tested exhibit chemorepulsion.

Mediation of opposing chemotactic responses by Tsr

We wondered if our inability to observe repulsion from fecal material might be due to *S. Typhimurium* not sensing indole as a chemorepellent, since this chemotactic response has only been previously described for *E. coli* (Table S1). We again employed CIRA to address this question, comparing chemotaxis responses to either 5 mM L-Ser or 5 mM indole, and found that *S. Typhimurium* responds rapidly to these two effectors as chemoattractants and chemorepellents, respectively (Fig. S2H-I). Treatment with 5 mM indole, a concentration at the upper end of what occurs in the human gut, induces rapid chemorepulsion with the bacteria vacating the region proximal to the source. Interestingly, the chemorepulsion response occurs faster than chemoattraction, with a zone of avoidance clearly visible within the first 10 s of indole exposure (Fig. S2I, Movie 4).

We next wondered if perhaps our fecal treatments contained insufficient indole to elicit chemorepulsion from *S. Typhimurium*. To identify the effective source concentrations that drive indole chemorepulsion and understand the temporal dynamics of this response, we performed a titration of indole across 0.05-10 mM. At all concentrations tested, indole induces chemorepulsion, and the bacteria avoid the treatment source for the duration of the 5-minute experiment. At source concentrations exceeding 3 mM, essentially all motile cells vacate the field of view within 60 s. Integrating these chemorepulsion responses and fitting them to a Monod curve suggests an indole source concentration of approximately 67 μ M is sufficient for half-maximal ($K_{1/2}$) chemorepulsion. These data show that even though we observe strong attraction to fecal material, pure indole at the concentration present in fecal material, and far lower, is indeed a strong chemorepellent for *S. Typhimurium*.

Based on its function in *E. coli*, we hypothesized that both indole chemorepulsion and L-Ser chemoattraction for *S. Typhimurium* could be partly or fully mediated through Tsr. We compared the chemotactic responses of the WT and *tsr* strains when exposed to sources of these effectors and found Tsr to be required for both chemorepulsion from indole and chemoattraction to L-Ser. The canonical mode of chemoreceptor effector recognition involves binding of the effector to the ligand-binding domain (LBD), but the mechanism by which indole is sensed through Tsr in *Salmonella* has not been elucidated. We recently reported the first crystal structure of *S. Typhimurium* Tsr LBD, which clearly defines how the binding site recognizes the L-Ser ligand, and we thought it unlikely indole can be accommodated at the same site. To our knowledge, no prior study has tested whether the Tsr LBD binds indole directly, so we expressed and purified the LBD, corresponding to the soluble periplasmic portion, and performed isothermal titration calorimetry (ITC). These data show that no binding occurs between the Tsr LBD and indole.

We next wondered if indole acts as an allosteric regulator of the LBD, possibly through interacting with the L-Ser-bound form or interfering with L-Ser recognition. To address these possibilities, we performed ITC of 50 μ M Tsr LBD with L-Ser in the presence of 500 μ M indole and observed a robust exothermic binding curve and K_D of 5 μ M, identical to the binding of L-Ser alone. These data indicate that indole does not alter the Tsr LBD affinity for L-Ser. We conclude that Tsr senses indole through an atypical mechanism, which might either involve regulation through a solute-binding protein, responsiveness to perturbation in the proton motor force, or binding to a different region other than the periplasmic LBD. Our

findings reveal that while indole acts as a chemorepellent for *S. Typhimurium* in isolation, sensed through Tsr, its presence within fecal material mixed with other effectors is insufficient to elicit chemorepulsion.

Compromising between conflicting effector signals through chemohalation

Earlier work with *E. coli* revealed that exposure to mixtures of L-Ser and indole can elicit intermediate chemotactic responses between chemoattraction and chemorepulsion after prolonged exposure (1-6 h, Table S1) ¹³. Having confirmed that Tsr in *S. Typhimurium* mediates opposing responses to the chemoattractant L-Ser and the chemorepellent indole in isolation, we next sought to learn how the bacterial population behaves on rapid timescales when confronted with physiological combinations of these effectors. To address this, we performed a series of CIRA experiments with 500 μ M L-Ser and increasing concentrations of indole at L-Ser:indole molar ratios of 10:1, 1:1, or 1:10 (Fig. 4A-D ¹³, Movie 4).

These experiments reveal a fascinating transition in the distribution of the pathogen population as a function of increasing chemorepellent, which occurs within minutes of exposure (Fig. 4A-D ¹³, Movie 4). With only chemoattractant present, the bacterial population organizes tightly around the effector source (Fig. 4A ¹³, Movie 4). When indole is introduced at a concentration 10-fold lower than L-Ser, the bacterial distribution still exhibits chemoattraction but becomes more diffuse (Fig. 4B ¹³, Movie 4). At a 1:1 ratio of chemoattractant and chemorepellent, a novel population structure emerges, in which the swimming bacteria are attracted toward the source but form a halo around the treatment with an interior region of avoidance (Fig. 4C ¹³, Fig. 4E ¹³, Movie 4). When the concentration of indole is 10-fold higher than L-Ser, the bacteria exhibit a wider zone of avoidance (Fig. 4D-E ¹³, Movie 4). Interestingly, whereas 5 mM indole on its own induces strong chemorepulsion (Fig. S2I, Movie 4), the addition of 10-fold lower L-Ser effectively converts the behavior to a null response (Fig. 4D-E ¹³, Movie 4). This demonstrates that even at the highest concentrations of indole *S. Typhimurium* might encounter in the gut, the presence of chemoattractant can override indole chemorepulsion.

The intermediate responses to opposing effector mixtures bear similarities to CIRA experiments with fecal material, some of which also exhibited a halo-like structure around the treatment source (Fig. 2 ¹³, Fig. 3A-E ¹³, Movies 2-3). As mentioned earlier, previous studies have also observed responses that are neither attraction nor repulsion, and they have been referred to by a variety of names. We do not claim here that the responses we observe to mixtures of L-Ser and indole are wholly distinct or unrelated to earlier work ^{13,18,19}, but to our knowledge, no other study has so clearly captured the spectrum of behaviors that occur in the transition between attraction and repulsion (Movie 4). There exists no consensus term for taxis of this nature, and so we suggest “chemohalation,” in reference to the halo formed by the cell population, and which is congruent with the commonly-used terms chemoattraction and chemorepulsion. We expect chemohalation is a compromise in positional location at the population level between the chemoattraction driven by L-Ser and the chemorepulsion driven by indole. Across these experiments, the interior zone of avoidance roughly corresponds to where the local concentration of indole exceeds 10 nM (Fig. 4E ¹³-F). In a biological setting, we presume that the distribution bias orchestrated by chemohalation regulates the probability that cells will colonize, adhere, and transition to sessility at a given site; the greater the local indole content, the wider the zone of avoidance and the less likely tissue invasion occurs.

We questioned why non-typhoidal *Salmonella* are attracted to a biological solution with high amounts of indole, a chemical reported to inhibit bacterial growth ^{21,66,67}. We examined how bacterial growth is affected by 0-25 mM indole or L-Ser in a background of minimal media (MM, Materials & Methods). As expected, increasing amounts of the nutrient L-Ser provide a growth advantage for all *Salmonella* strains analyzed, with maximal benefit achieved by

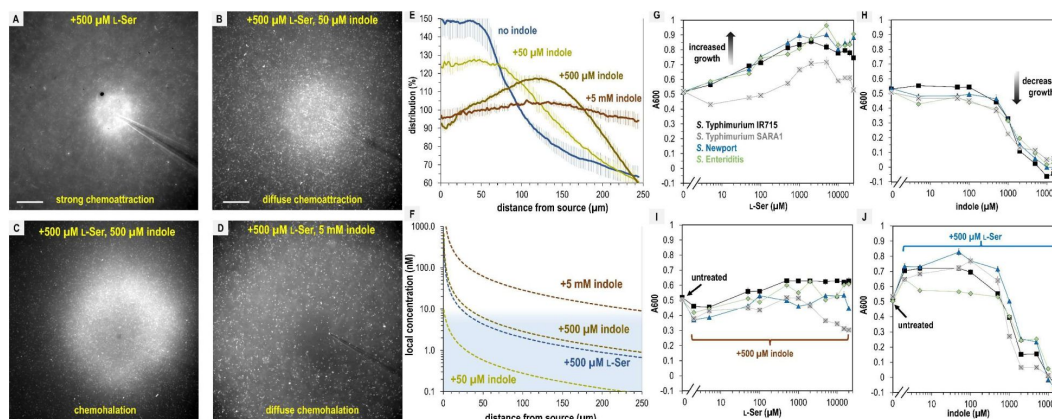


Fig. 4.

***S. Typhimurium* mediates distinct chemotactic responses based on the ratio of L-Ser to indole.**

A-D. Representative max projections of responses to treatments of L-Ser and indole at 295-300 s, as indicated. Scale bars are 100 μ m. E. Relative bacterial distribution in response to treatments of 500 μ M L-Ser and varying amounts of indole, from panels A-D, with the mean value normalized to 100%. Data are means and error bars are standard error of the mean (SEM, n=3-5). F. Diffusion modeling of local effector concentrations based on sources of 5 mM indole (dark brown), 500 μ M L-Ser (blue), 500 μ M indole (light brown), and 50 μ M indole (yellow) are shown as dashed lines. The approximate local concentration of indole that elicits a transition in chemotactic behavior is highlighted in light blue. G-H. Bacterial growth as a function of L-Ser or indole, at the time point where the untreated culture reaches A600 of 0.5. I-J. Bacterial growth +/- pretreatment with 500 μ M indole or L-Ser, and increasing concentrations of indole or L-Ser, as indicated at the time point where the untreated culture reaches A600 of 0.5. Data are means and error bars are standard error of the mean (SEM, n=8-24).

approximately 500 μ M (**Fig. 4G**). Equivalent treatments with indole show tolerance up to approximately 1 mM, with growth inhibition occurring in the 1-5 mM range and lethality occurring at indole concentrations greater than 5 mM (**Fig. 4H**). However, adding L-Ser in a background of 500 μ M indole provides only a small growth enhancement (**Fig. 4I**), and addition of 500 μ M L-Ser increases tolerance for indole up to about 1 mM, above which indole toxicity is unavoidable (**Fig. 4J**). So, we conclude that mixtures of these effectors also impact growth differently than the effectors in isolation, and the relative attraction to combinations of these effectors relates to their propensity to enhance or inhibit growth. Overall, the bacteria still obtain growth benefits from L-Ser so long as the concentration of indole is under 1 mM, and we expect this growth advantage is even greater in the presence of the many other nutrients accessible to the bacteria in fecal material.

Discussion

Bacteria in the human gastrointestinal tract encounter complex chemical landscapes that contain both chemoattractants and chemorepellents. However, chemotaxis responses are often studied in isolation, outside of their biological and ecological contexts, which can lead to an over- or underestimation of the roles specific interactions play in natural settings. In the present work, we contribute to an emerging understanding that bacteria exhibit rapid and well-orchestrated responses to conflicting stimuli distinct from chemoattraction or chemorepulsion, and relate these chemotactic compromises to enteric infection and pathogen growth (**Fig. 5**). In this study, we provide evidence that despite the microbiota metabolite indole being a strong repellent in isolation, fecal indole is insufficient to elicit pathogen repulsion or protect against invasion.

New insights into indole taxis from a non-*E. coli* system

Indole is an important and abundant regulator of enteric microbial communities known to modulate motility and virulence and has been proposed to deter pathogen invasion through chemorepulsion. Previously, no study had addressed whether bacteria other than *E. coli* sense indole as a chemorepellent. In the model system we investigated, we confirm that *S. Typhimurium* utilizes the chemoreceptor Tsr to respond to indole as a chemorepellent (**Fig. 3**). Further, we add to the understanding of indole taxis by having visualized and measured the rapid temporal dimension of indole repulsion, which is relevant in the highly turbulent intestinal environment where bacteria experience shifting and dynamic chemical gradients. We found naïve *S. Typhimurium* are repelled from indole sources within 10 s of exposure. The rapidity of this response, which exceeds that of attraction to L-Ser, has not been previously reported (**Fig. 3G**). We also tested whether indole-sensing occurs through the canonical chemoreceptor mechanism of direct binding through the Tsr LBD, which does not seem to occur, nor does antagonism or inhibition of binding of L-Ser to the LBD (**Fig. 3L-M**). These data support that a non-canonical sensing mechanism is employed by Tsr to respond to indole, as suggested by others.

Having characterized and confirmed the role of Tsr in indole-sensing for *S. Typhimurium*, and shown it to be generally similar to its function in *E. coli*, we can speculate that the diverse bacterial species that possess Tsr orthologues, particularly common among *Enterobacteriaceae*, also sense, and are repelled, by indole, further supporting indole as a key regulator of polymicrobial communities of the gut. However, *in vivo* Tsr-mediated indole-sensing is integrated with L-Ser-sensing and responses to other stimuli. As discussed further below, this integration may serve to bias colonization toward specific niches based on the local attractant-to-repellent ratio, rather than triggering the outright repulsion we observe *in vitro* with pure indole sources (**Fig. 5**). While Tsr appears to function similarly in coordinating indole taxis in

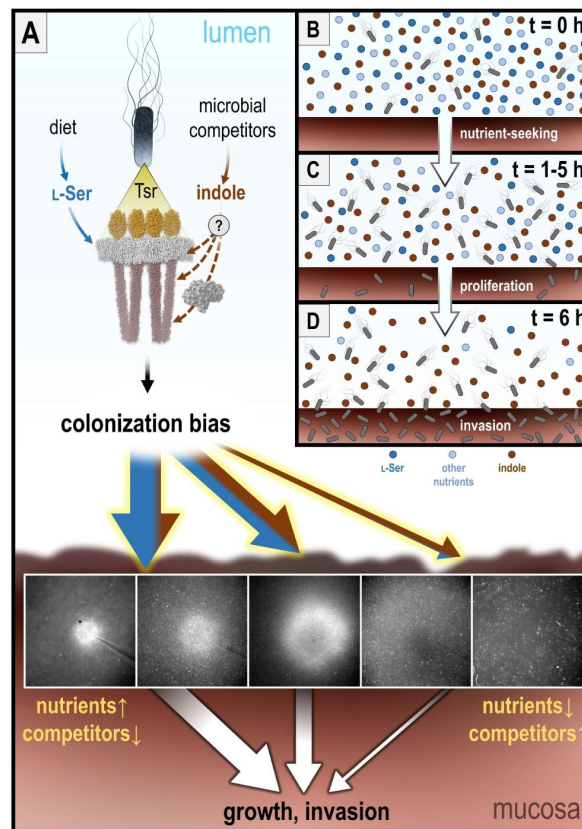


Fig. 5.

A. Working model for chemotactic compromise orchestrated by Tsr during enteric invasion.

In the intestinal lumen, fecal material contains the nutrient L-Ser, derived from digested food, and bacteriostatic indole, secreted by the gut microbiota. Tsr detects these metabolites as chemoattractant (L-Ser, dark blue arrows) and chemorepellent (indole, brown arrows) signals, respectively. While L-Ser is directly sensed by Tsr via the ligand-binding domain (LBD, yellow), indole is detected through a different mechanism. Tsr integrates these conflicting signals to bias colonization through a behavior termed chemohalation, which promotes taxis toward regions with higher L-Ser-to-indole ratios (large blue and brown arrows, width of arrows denotes the degree of colonization bias). Increased indole levels result in a larger central zone of avoidance and greater bias against colonization. B. During early infection, the attractant-to-repellent ratio in the intestinal lumen is high. *S. Typhimurium* (gray cells) uses chemotaxis to seek L-Ser (dark blue circles) and other nutrients (light blue circles), promoting proliferation. In the presence of abundant nutrients, the bacteria tolerate exposure to indole (brown circles). C. As infection progresses, nutrient consumption by the proliferating bacteria reduces the attractant-to-repellent ratio in the lumen. D. Once nutrients are sufficiently depleted, the bacteria can no longer tolerate indole exposure, and chemorepulsion from indole increases. This repulsion drives bacteria out of the lumen and into contact with the intestinal mucosa, promoting colonic invasion. In the artificial scenario where only L-Ser is present, chemotaxis offers little advantage for invasion; the bacteria proliferate, but there is no incentive to leave the lumen. Conversely, with only indole present, chemotaxis similarly provides no advantage; although indole repels the bacteria from the lumen, it inhibits bacterial proliferation. In the presence of L-Ser, and likely other attractants, indole repulsion is nullified.

both *S. Typhimurium* and *E. coli*, its impact on colonization is likely different. Unlike *Salmonella*, *E. coli* produces indole^{39,40}, which may allow it to use indole not only to regulate colonization based on microbiota activity but also as a self-sensing signal. This could enable *E. coli* to regulate its dispersal or expansion in response to self-generated indole.

Indole taxis amidst opposing stimuli and implications for pathogen invasion

We investigated the chemotaxis behaviors of *S. Typhimurium* to human fecal material as a test case for how pathogens make decisions when faced with biological sources of effectors that contain a mixture of conflicting stimuli (Fig. 1, Fig. 2). We predicted that feces, rich in indole, would elicit chemorepulsion, inhibit pathogen growth, and protect against infection^{12,20–22,25,28,29}. Instead, we find that chemotactic sensing of human fecal material promotes colonic invasion in an explant model, and liquid feces predominantly elicits chemoattraction or chemohalation (Fig. 1, Fig. S1, Fig. 2, Movies 1–S). We also find that while both invasion and chemotaxis are regulated by the presence of the fecal effectors indole or L-Ser alone, these effectors in combination cancel out, leading to a null response (Fig. 1, Fig. 4). These findings help clarify why indole-rich fecal material does not trigger repulsion or prevent invasion. While indole is abundant in feces, it is only one component within a complex biological mixture, and other effectors ultimately mediate chemoattraction or chemohalation to fecal material (Fig. 5).

To better understand how chemotaxis responses change when sensing mixtures of opposing effectors, we examined bacterial behavior *in vitro* in the presence of physiological mixtures of the fecal metabolites indole and L-Ser and captured a series of real-time videos (Fig. 4A–D, Movie 4). These videos reveal how, upon sensing the conflicting stimuli, the pathogen population structure rapidly evolves based on the ratio of attractant to repellent, ranging from chemoattraction, diffuse chemoattraction, chemohalation, diffuse chemohalation, and chemorepulsion (Fig. 4A–D, Fig. 5, Fig. S2, Movie 4). We found particularly interesting the response we refer to as chemohalation, which occurs when the bacteria encounter near equimolar attractant and repellent. Under these conditions, the cells accumulate at a distance from the treatment source and form a halo with an interior zone of avoidance (Fig. 3, Movie 4). The zone of avoidance apparently increases as a function of increasing indole, thereby biasing cells against colonizing and invading a specific niche (Fig. 5A). Whereas we observe chemohalation *in vitro* with only two fecal metabolites, we expect this mechanism is at play when the bacteria are selecting among niches in the intestinal lumen, where the indole repulsion signal is integrated alongside information about nutrient attractant signals to ultimately coordinate contact with, and invasion of, intestinal tissue (Fig. 5A).

Potential roles for chemohalation in gastrointestinal infections

The dynamic micron-scale chemohalation population structures, that in effect are a compromise between attraction and repulsion, would be difficult or impossible to detect without directly viewing them through live imaging, which may explain why they were unappreciated earlier in binary models of chemotaxis¹⁵. To be clear, we suggest chemohalation as a useful term to generally describe intermediate chemotaxis responses to conflicting stimuli that are neither chemoattraction nor chemorepulsion, but others have also contributed to studying how chemotaxis functions in confounding chemical landscapes^{13,18,19}. The chemohalation responses we report here are most similar to what was referred to as a “trade-off” response in a prior study with *E. coli* and attractant-repellent mixtures; interestingly those authors also observed another response termed “bet-hedging” where a portion of the population within an isogenic culture remained attracted despite the presence of the chemorepellent, but we did not observe this behavior in our experiments¹³. For *S. Typhimurium*, the source concentration required for half-maximal responses are similar for the opposing effectors, 105 and 67 μM for L-Ser¹ and indole, respectively, but the presence of other attractants in feces overall biases behavior toward attraction^{57–59,61}.

In the context of non-typhoidal *Salmonella* infections, it is clear there exists complex relationships between chemotactic sensing of effectors, bacterial growth, and invasion. As it pertains specifically to sensing the opposing effectors L-Ser and indole, we propose that Tsr functions to measure the ratio of exogenous attractant to repellent, and, through chemohalation, orchestrates an appropriate spatial bias in the population. Ultimately, this serves to improve pathogen fitness through colonizing niches rich in nutrients, signaled by local L-Ser concentrations, while avoiding niches with high microbial competition, indicated by local indole concentrations (Fig. 5). Chemorepulsion from indole can be overridden by the presence of chemoattractants, and *S. Typhimurium* growth is quite tolerant of indole within physiological ranges, suggesting the bacteria generally prioritize nutrient acquisition over the inhibitory effects of indole (Fig. 4). This agrees with the premise that nutrient-seeking, rather than avoidance of toxins, may generally be the major function of chemotaxis^{6,18}. We expect that the distinction between chemoattraction and chemohalation, in terms of regulating colonization and infection outcomes, also depends on the duration of stimuli exposure and adaptation¹². Based on data from colonic explant experiments, chemotaxis studies, and growth analyses, we propose a new model describing how the opposing signals from L-Ser and indole are sensed and integrated by Tsr to mediate pathogen invasion (Fig. 5). In addition to sensing of these two effectors, our data support that Tsr and chemotaxis are involved in L-Ser and indole-independent functions in the presence of fecal material that mediate colonization (Fig. 1). This could potentially relate to other characterized functions of Tsr in redox and energy taxis and/or biasing swimming toward a running phenotype, which is associated with increased invasion^{33,34,38}.

Recently, we reported on *Enterobacteriaceae* chemotactic sensing of blood serum, another complex biological effector source at the host-pathogen interface, and those responses appear to involve chemohalation¹. Evidence of chemohalation is also seen in the case of the gastric pathogen *Helicobacter pylori* responding to mixtures of urea, a chemoattractant, and acid, a chemorepellent^{19,69}. Continuing to investigate chemohalation behaviors and understanding how they coordinate bacterial colonization may provide important insights into how chemotaxis confers fitness advantages in natural environments.

Limitations of this study

This study provides insights into the roles of chemotaxis in *Salmonella* responses to fecal material and indole; however, several limitations should be considered. First, our experiments were conducted *in vitro* and utilized swine tissue explants, which do not fully capture the complexities of *in vivo* infection dynamics in the human gut. Additionally, while explant assays provided insight into how chemotaxis relates to tissue colonization, these experiments exhibited variability. To mitigate this, we used multiple sections of tissue from a single animal to exert greater experimental control of the system, but this approach limits our ability to assess how host tissue diversity might influence bacterial responses. It would be interesting to assess invasion using tissue of the distal ileum, a major site of *S. Typhimurium* invasion, which is known to have distinct chemical cues and foreseeably differences in indole and L-Ser⁴¹. Another experimental limitation is the difference in timescales between our assays. Chemotaxis experiments were conducted over short time frames, whereas tissue explant experiments required longer incubation periods for significant differences in colonization and invasion to be observed. Lastly, while we confirmed that non-typhoidal *Salmonella* are attracted to human fecal material, we only determined the dependency on Tsr in our model strain (IR715). Although we expect that other WT clinical isolates also rely on chemotaxis and Tsr for attraction, this remains uncertain without targeted genetic analyses in each strain background.

Materials & Methods

All methods were carried out in accordance with relevant guidelines, regulations, and state and federal law. Experimental protocols were approved by the Institutional Biosafety Committee (IBC) of Washington State University (#1372).

Bacterial strains and growth conditions

Bacterial strains and plasmids used in this study are listed in [Table 1](#). As previously described [1](#), bacteria intended for chemotaxis assays were grown overnight in tryptone broth (TB) with antibiotic selection, as appropriate. Motile bacteria were prepared with a 1:1000 back-dilution and grown shaking for approximately 4 hours at 37°C to reach A_{600} of 0.5. Cells were centrifuged, washed, and resuspended in a chemotaxis buffer (CB) containing 10 mM potassium phosphate (pH 7), 10 mM sodium lactate, and 100 μ M EDTA to A_{600} of 0.2 and rocked gently at room temperature until fully motile. For *in vitro* growth analyses, cultures were grown overnight in Lysogeny Broth (LB) at 37°C. Subsequently, 5 μ l of A_{600} 2.0 cells were used to inoculate 200 μ l of minimal media (MM), containing 47 mM Na_2HPO_4 , 22 mM KH_2PO_4 , 8 mM NaCl, 2 mM MgSO_4 , 0.4% glucose (w/v) 11.35 mM $(\text{NH}_4)_2\text{SO}_4$, 100 μ M CaCl_2 and L-Ser and/or indole at the described concentrations, and cultured in a 96-well microtiter plate. Cultures were grown at 37°C and monitored by A_{600} readings at 5-minute intervals.

Chemosensory injection rig assay (CIRA)

CIRA was performed as described previously [1](#). Briefly, an Eppendorf Femtotip 2 microcapillary containing the treatment of interest was lowered into a pond of 50 μ l of motile cells using a Sutter micromanipulator. An injection flow of effector into the pond at approximately 300 fl per minute was achieved using a Femtojet 4i set to P_c 35. Solubilized fecal treatments were prepared by dissolving 1 g of commercially obtained human feces (Lee Biosolutions) in 10 ml of CB. The solution was clarified by centrifugation at 10,000 g for 20 minutes, followed by sterile filtration through a 0.2 μ m filter. Treatment solutions of indole and L-Ser were also diluted into CB and sterile-filtered before application. Videos were recorded using an inverted Nikon Ti2 microscope with heated sample chamber at 37 °C.

CIRA microgradient modeling

Modeling the microgradient generated through CIRA was performed as described earlier [1](#), based on the continual injection and diffusion of an effector from a fixed-point source. Briefly, diffusion is modeled as a 3D process where the diffusing substance is gradually and continuously introduced at a fixed point within a large surrounding fluid volume. The substance is prepared at a concentration of M_s (typically between 0.5 μ M and 5 mM) and injected at a volume rate of $Q = 305.5$ fl/min. The species then diffuses into the ambient fluid with a diffusion constant D :

$$C(r, t) = \frac{q}{4\pi Dr} \operatorname{erfc} \frac{r}{2\sqrt{Dt}}$$

Here, r is the distance from the point source, t is the time from initial injections, q is the injection rate of the species (equal to $M_s Q$), and C is the species concentration. In earlier work [1](#), we reported using fluorescent dye that the concentrations predicted by this model appear to be accurate within 5% in the range of 70-270 μ m from the source, whereas at distances less than 70 μ m the measured concentrations are about 10% lower than predicted. At the point where the effector treatment is injected into the larger volume the local concentration drops precipitously, hence why the concentration reported at distance 0 is not that of the concentration within the microcapillary.

Isothermal titration calorimetry ligand binding studies (ITC)

Purification of *S. Typhimurium* Tsr LBD was performed as described previously¹. ITC experiments were performed using a Microcal ITC200 instrument (GE Healthcare). Either 500 μ M indole or L-Ser was titrated in 2.5 μ L injections into a 200 μ L sample cell containing 50 μ M Tsr LBD. For the indole/L-Ser competition experiment, 500 μ M indole was added to both the titrant and sample cell, thus providing a constant excess background concentration of indole. For all experimental conditions, blank titrations were also collected in which indole or L-Ser was titrated into a cell containing buffer alone. All experiments were performed using thoroughly degassed samples at 25 °C in 50 mM Tris, 150 mM NaCl, 1 mM EDTA, pH 7.5. The reference power was set to 5 μ cal/sec. The resulting power curves were integrated using the Origin analysis software included with the instrument. The heat of dilution was subtracted from each point using the blank. A single-site binding model was then fit to the data, floating parameters describing the binding enthalpy (ΔH), equilibrium constant (K_D), and apparent binding stoichiometry (n). The instrument software was used for this purpose.

Quantification of indole and serine in human fecal samples

Solubilized human feces was prepared as described above for CIRA and analyzed by mass spectrometry to determine the molar serine content as a service through the University of Washington Mass Spectrometry Center. This measurement reflects total serine, of which close to 100% is expected to be L-Ser¹. As described in earlier work, the indole content of solubilized human fecal samples was determined using a hydroxylamine-based calorimetric assay with purified indole as a reference and standard⁷².

Explant infection assays

Swine intestinal tissue was acquired from the descending colon of an 8-week-old animal, pursuant to animal protocol ASAF #7128, approved through the Washington State University IACUC. Before infection, the luminal side of an approximately 20 by 20 mm piece of swine intestinal explant tissue was gently washed with PBS to remove fecal matter. Next, the tissue section was bathed in 2 ml of chemoeffector solution (solubilized human fecal matter (Lee Biosolutions), a mixture of 338 μ M L-Ser and 862 μ M indole, 338 μ M L-Ser alone, 862 μ M indole alone, or chemotaxis buffer) in a 6-well tissue culture plate (Celltreat) and incubated at 4°C for 1 h. Then, tissue was transferred to a 35 mm Mattek dish where the luminal side of the tissue was exposed to a bacterial solution containing a 1:1 mixture (10^9 CFU each) of WT *S. Typhimurium* IR715 and either the isogenic *tsr* or *cheY* mutant, suspended in CB at a volume of 300 μ L. The tissue was then incubated in the dish with the competing bacteria at 37 °C and 5% CO₂ for 1, 3, or 6 h. After, half of the tissue was transferred into screwcap tubes containing 500 μ L LB media and 5-10 2.3 mm zirconia beads (BioSpec Products) on ice and homogenized using a Bead Mill 24 (Fisher Scientific) at 6.5 m/s for 60 s, repeated four times. To enumerate the intracellular bacteria, the other half of the tissue was washed in PBS and incubated in PBS containing 100 μ g/ml gentamicin for 1 h at 37 °C and 5% CO₂, then washed twice in PBS, as done previously^{54,73,74}. The homogenization process was then repeated for the gentamicin-treated tissue. CFUs were enumerated by plating 10-fold dilutions on LB agar plates containing the appropriate antibiotic^{54,75}. Competitive index values were calculated by dividing the number of mutant CFUs by the number of WT CFUs for each treatment and time point^{76,77}.

Quantification of CIRA data

Videos of chemotactic responses were quantified as described previously¹. The number of cells in each frame were calculated by determining a fluorescence intensity ratio per cell for frames pre-treatment and extrapolated using the ‘plot profile’ function of ImageJ. The distribution of the

bacteria was calculated using the Radial Profile ImageJ plugin. Local background subtraction was performed based on experiments with the non-chemotactic *cheY* strain to control for autofluorescence in solubilized fecal samples.

Statistical Analyses

Competitive indices (CIs) for explant experiments were calculated for each treatment group at each time point. Log-transformed CI values were obtained by taking the logarithm ($\log_{10}$) of the original CI measurements. These log-transformed values were then subjected to statistical analysis. First, a one-sample t-test was performed to determine whether the mean of the log-transformed CIs significantly differed from zero. In cases where the assumption of normality was violated, the non-parametric Wilcoxon rank sum test was applied as an alternative. Effect size was assessed using Cohen's *d* and calculated using the same log-transformed CIs.

Supplementary tables and figures

Species, Strain(s)	Indole Treatment ^a	Reported Response ^b	Experimental Method(s)	Chemotaxis & Motility Proteins Involved in Response	Ref.
<i>E. coli</i> , strain RP437	≤1 mM >1 mM	chemorepulsion chemoattraction	tethered cell assay ^c	Tsr, Tar	¹²
<i>E. coli</i> , strain MG1655	1 mM ± 0.5- 0.1 mM L-Ser	“bet-hedging” ^d	diffusion-based assay into long channel with fluorescent imaging	Tar, Tsr	¹³
<i>E. coli</i> , strain RP437	500 μM	chemorepulsion, overridden in presence of 500 μM Autoinducer-2	flow-based microfluidic chemotaxis device coupled to a gradient generator	Tar, Tsr	¹⁸
<i>E. coli</i> , strain RP437	≤1 mM >1 mM	decreased motility slower flagellar rotation	tethered cell assay ^c	Non-CheY dependent	²²
<i>E. coli</i> , strain O157:H7 CDC EDL933	500 μM	chemorepulsion	agarose plug chemotaxis assay imaged every 5 minutes for 30 minutes	MotB and FlhD	²⁷
<i>S. Typhimurium</i> , strain ATCC14028s	1 mM	decreased motility	swimming motility agar plates ^c	MotA	²⁸
<i>S. Typhimurium</i> , strain ATCC14028s	1 mM	decreased motility	swimming motility agar plates ^c	FlhC	²⁹

Table S1.

Summary of prior studies related to indole chemotaxis, related to Figure 1 [↗](#).

^a Source concentration, or ranges, used in experiment.

^b The motility or chemotactic response as reported by the study authors; note that some methods employed may not be able to distinguish between responses as a consequence bacterial growth versus chemotaxis.

^c The assay may have limitations in its ability to report on either rapid temporal responses, or localization to or from an effector source.

^d In this work, we refer to behaviors of this type as “chemohalation” to be similar to the widely-used terms chemoattraction and chemorepulsion.

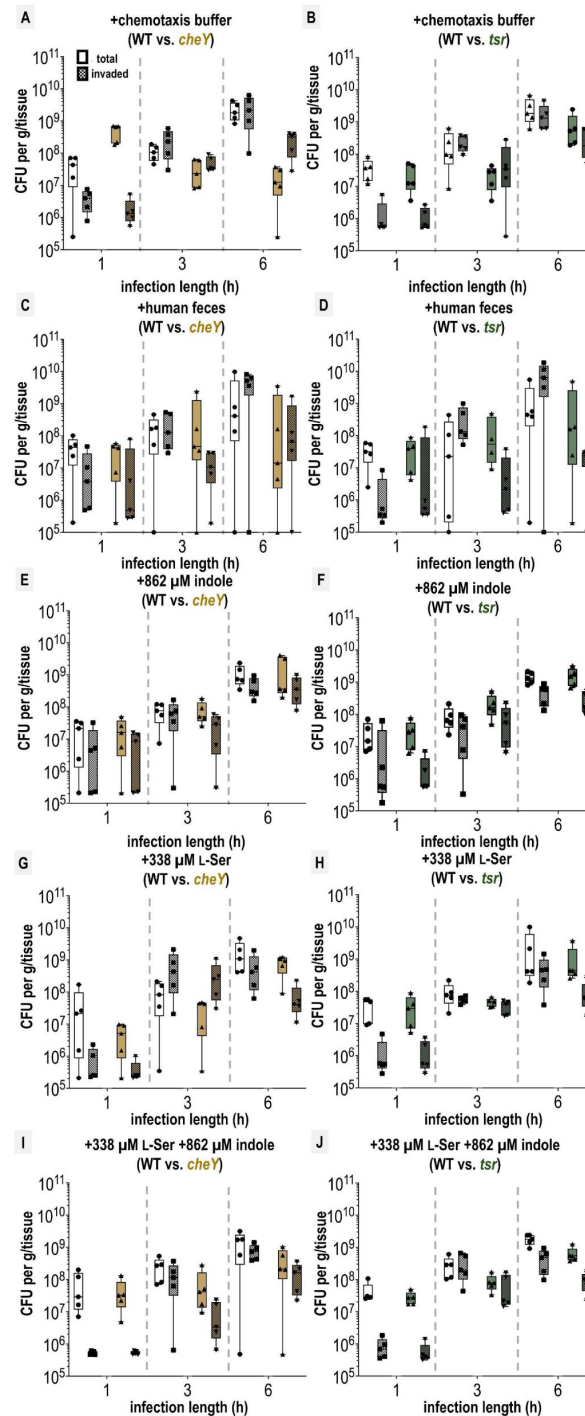


Fig. S1.

Colony-forming units (CFU) recovered from swine colonic explant infections, related to Figure 1

Shown are disaggregated CFU enumerations for each co-infection experiment with *S. Typhimurium* IR715 WT and mutant strains following tissue treatments, as indicated. CFUs extracted either using the entire tissue homogenate (total) or extracted from tissue treated with a gentamicin wash to kill exterior non-invaded cells (invaded) are noted separately (see Materials & Methods). Each data point represents CFUs recovered from a single explant experiment ($n=5$). Box and whisker plots represent the sample median (line), the edges are the upper and lower quartiles, and whiskers extend to the max and min values. The limit of detection was 1×10^5 CFUs. Samples with no CFUs detected are noted on the plot at this limit.

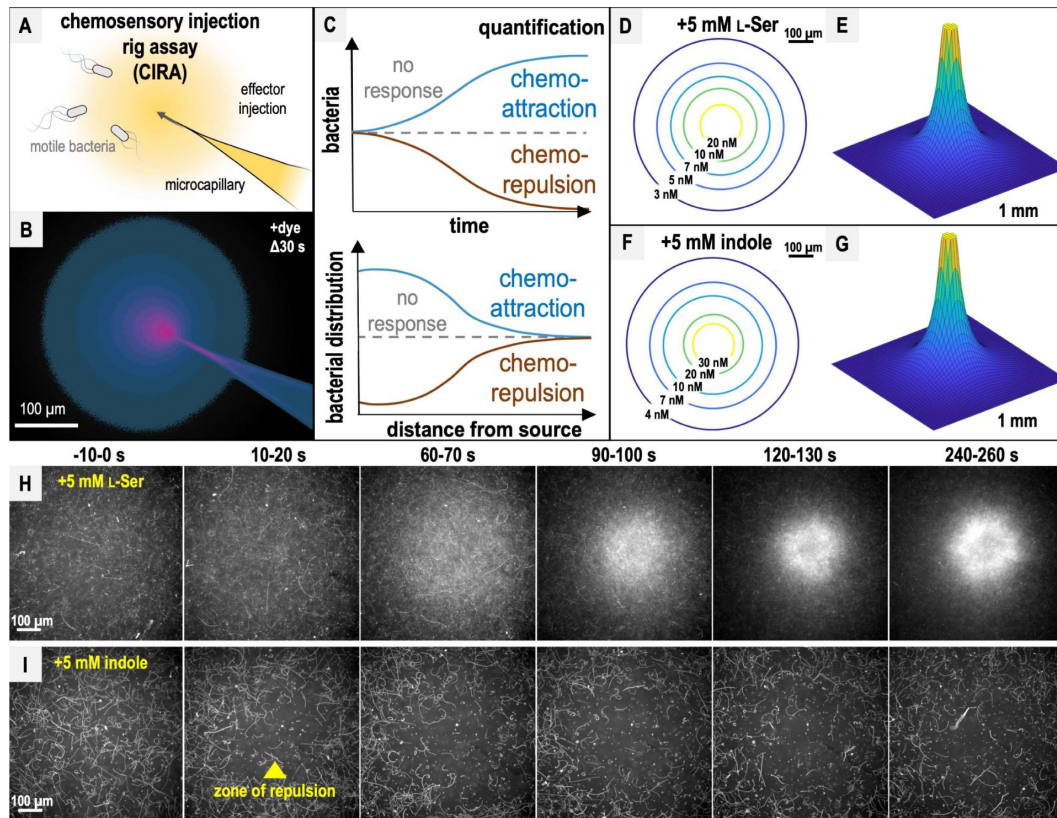


Fig. S2.

Chemotactic responses to L-Ser or indole, related to Figure 2 [↗](#).

A. CIRA experimental design. B. CIRA microgradient diffusion model, simulated with a source of 1.13 mM A488 dye after 30 s of injection. Image rendered on a pink (100% source concentration)-blue-black (0%) color intensity scale, based on previous work [1](#). C. CIRA quantification overview, with blue lines corresponding to chemoattraction, null chemotactic response corresponding to gray dashed lines, and chemorepulsion corresponding to brown lines. D-G. Microgradient models of the local effector concentration for treatments with 5 mM L-Ser or indole, respectively, at $\Delta 300$ s. Color gradient corresponds to gradually decreasing effector concentrations from yellow to blue. H-I. Representative 10 s max projections of *S. Typhimurium* IR715 response to sources of L-Ser (strong chemoattraction) or indole (strong chemorepulsion) using CIRA. The glass microcapillary that injects the treatment solution is centered within the field of view. Note that cells contain a plasmid expressing mPlum (see Materials & Methods), and fluorescence data are collected in the far-red channel, so the glass microcapillary is poorly visible.

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Additional information

Author Contributions

K.F. performed the microgradient modeling and explant experiments. A.B. conducted the CIRA experiments and purification of Tsr-LBD. Z.G. performed bacterial growth experiments. M.S. and

M.J.H. performed the ITC experiments. All authors contributed to data analyses and writing of the manuscript.

Additional files

Data S1. Enumeration of colony-forming units (CFUs) from explant studies. [🔗](#)

Movie 1. Chemotactic response of *S. Typhimurium* IR715 to solubilized human feces, related to Figure 2. Representative CIRA experiments showing *S. Typhimurium* IR715 WT and mutant strains responding to a source over 300 s (shown at 10x speed) [🔗](#)

Movie 2. Chemotactic response of *S. Typhimurium* IR715 WT and chemotactic mutant strains to solubilized human feces, related to Figure 3. Representative CIRA experiments showing competition between *S. Typhimurium* IR715 (mPlum) and *cheY*, or *tsr*, as indicated (GFP), over 300 s. [🔗](#)

Movie 3. Chemotactic response of *S. enterica* clinical isolates to solubilized human feces, related to Figure 3. Representative CIRA experiments showing competition between *S. Typhimurium* IR715 (mPlum) and clinical isolates, as indicated (GFP), responding to a source of solubilized human feces over 300 s. [🔗](#)

Movie 4. Chemotactic response of *S. Typhimurium* IR715 to L-Ser and indole treatments, related to Figure 4 and Figure S2. Representative CIRA experiments with treatment sources as indicated, over 300 s. [🔗](#)

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Reviewer #1 (Public review):

Summary:

The study shows, perhaps surprisingly, that human fecal homogenates enhance the invasiveness of *Salmonella typhimurium* into cells of a swine colonic explant. This effect is only seen with chemotactic cells that express the chemoreceptor Tsr. However, two molecules sensed by Tsr that are present at significant concentrations in the fecal homogenates, the repellent indole and the attractant serine, do not, either by themselves or together at the concentrations in which they are present in the fecal homogenates, show this same effect. The authors then go on to study the conflicting repellent response to indole and attractant response to serine in a number of different in vitro assays.

Strengths:

The demonstration that homogenates of human feces enhance the invasiveness of chemotactic *Salmonella Typhimurium* in a colonic explant is unexpected and interesting. The authors then go on to document the conflicting responses to the repellent indole and the attractant serine, both sensed by the Tsr chemoreceptor, as a function of their relative concentration and the spatial distribution of gradients.

Weaknesses:

The authors do not identify what is the critical compound or combination of compounds in the fecal homogenate that gives the reported response of increased invasiveness. They show it is not indole alone, serine alone, or both in combination that have this effect, although both are sensed by Tsr and both are present in the fecal homogenates. Some of the responses to conflicting stimuli by indole and serine in the in vitro experiments yield interesting results, but they do little to explain the initial interesting observation that fecal homogenates enhance invasiveness.

<https://doi.org/10.7554/eLife.106261.1.sa3>

Reviewer #2 (Public review):

Summary:

The manuscript presents experiments using an ex vivo colonic tissue assay, clearly showing that fecal material promotes *Salmonella* cell invasion into the tissue. It also shows that serine and indole can modulate the invasion, although their effects are much smaller. In addition, the authors characterized the direct chemotactic responses of these cells to serine and indole using a capillary assay, demonstrating repellent and attractant responses elicited by indole and serine, respectively, and that serine can dominate when both are present. These behaviors are generally consistent with those observed in *E. coli*, as well as with the observed effects on cell invasion.

Strengths:

The most compelling finding reported here is the strong influence of fecal material on cell invasion. Also, the local and time-resolved capillary assay provides a new perspective on the cell's responses.

Weaknesses:

The weakness is that indole and serine chemotaxis does not seem to control the fecal-mediated cell invasion and thus the underlying cause of this effect remains unclear.

In addition, the fact that serine alone, which clearly acts as a strong attractant, did not affect cell invasion (compared to buffer) is somewhat puzzling. Additionally, wild-type cells showed nearly a tenfold advantage even without any ligand (in buffer), suggesting that factors other than chemotaxis might control cell invasion in this assay, particularly in the serine and indole conditions. These observations should probably be discussed.

Final comment. As shown in reference 12, Tar mediates attractant responses to indole, which appear to be absent here (Figure 3J). Is it clear why? Could it be related to receptor expression?

<https://doi.org/10.7554/eLife.106261.1.sa2>

Reviewer #3 (Public review):**Summary:**

In this manuscript, Franco and colleagues describe careful analyses of *Salmonella* chemotactic behavior in the presence of conflicting environmental stimuli. By doing so, the authors describe that this human pathogen integrates signals from a chemoattractant and a chemorepellent into an intermediate "chemohalation" phenotype.

Strengths:

The study was clearly well-designed and well-executed. The methods used are appropriate and powerful. The manuscript is very well written and the analyses are sound. This is an interesting area of research and this work is a positive contribution to the field.

Weaknesses:

Although the authors do a great job in discussing their data and the observed bacterial behavior through the lens of chemoattraction and chemorepulsion to serine and indole specifically, the manuscript lacks, to some extent, a deeper discussion on how other effectors may play a role in this phenomenon. Specifically, many other compounds in the mammalian gut are known to exhibit bioactivity against *Salmonella*. This includes compounds with antibacterial activity, chemoattractants, chemorepellents, and chemical cues that control the expression of invasion genes. Therefore, authors should be careful when making conclusions regarding the effect of these 2 compounds on invasive behavior. It is important that the word invasion is used in the manuscript only in its strictest sense, the ability displayed by *Salmonella* to enter non-phagocytic host cells. With that in mind, authors should discuss how other signals that feed into the control of *Salmonella* invasion can be at play here.

<https://doi.org/10.7554/eLife.106261.1.sa1>

Author response:

We thank the reviewers of this manuscript for their thoughtful and detailed feedback, and agree that they bring up valid points. We also thank them for their suggestions on how to improve this study. We intend to revise this manuscript to help address these concerns and in the future will submit a revised version that will hopefully be improved in terms of the clarity of the text and rigor of the experimental findings.

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